Algorithm 2 Region-Based Multicast Routing in the Upper Mesh

Require: $R_m = [x_L, x_H] \times [y_L, y_H]$, router (x, y), ingress d_{in}
Ensure: $\mathbf{D} = [D_W, D_S, D_E, D_N, D_L]$

$\mathbf{D} \leftarrow \mathbf{0}$
if $(x, y) \notin R_m$ **then**
  $C(R_m) \leftarrow \{(x_L, y_L), (x_L, y_H), (x_H, y_L), (x_H, y_H)\}$
  $(x_t, y_t) \leftarrow \arg\min_{(a,b) \in C(R_m)} |x - a| + |y - b|$
  if $x \neq x_t$ **then**
    $D_E \leftarrow (x < x_t)$; $D_W \leftarrow (x > x_t)$
  else
    $D_N \leftarrow (y < y_t)$; $D_S \leftarrow (y > y_t)$
  end if
else
  $D_L \leftarrow (d_{\text{in}} \neq L)$
  if $d_{\text{in}} = W$ **then**
    $D_E \leftarrow (x < x_H)$; $D_N \leftarrow (y < y_H)$; $D_S \leftarrow (y > y_L)$
  else if $d_{\text{in}} = E$ **then**
    $D_W \leftarrow (x > x_L)$; $D_N \leftarrow (y < y_H)$; $D_S \leftarrow (y > y_L)$
  else if $d_{\text{in}} = N$ **then**
    $D_S \leftarrow (y > y_L)$; $D_W \leftarrow (y = y_H \wedge x > x_L)$
    $D_E \leftarrow (y = y_H \wedge x < x_H)$
  else if $d_{\text{in}} = S$ **then**
    $D_N \leftarrow (y < y_H)$; $D_W \leftarrow (y = y_L \wedge x > x_L)$
    $D_E \leftarrow (y = y_L \wedge x < x_H)$
  else ▷ Local injection
    $D_W \leftarrow (x > x_L)$; $D_E \leftarrow (x < x_H)$
    $D_S \leftarrow (y > y_L)$; $D_N \leftarrow (y < y_H)$
  end if
end if
return D